

Aditya Vasu Rana

Chennai, India | adityaranaaa3113@gmail.com | 7876007402 | linkedin.com/in/aditya-rana | github.com/adityaa-rana

Education

Vellore Institute of Technology, Chennai

Aug 2023 – May 2027

Bachelor of Technology (B.Tech) in Computer Science, **CGPA : 9.5**

- **Coursework:** Computer Networks, Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Software Engineering, Database Management Systems

Technical Skills

- **Programming Languages:** Python, R, SQL, SAS, C, C++
- **Data Analysis & Modeling:** Statistical Modeling, Machine Learning, Regression, Clustering, Neural Networks
- **Tools & Platforms:** Git, AWS, Docker, Jupyter, Streamlit
- **Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Seaborn
- **Databases & Visualization:** SQL, PL/SQL, Tableau, Power BI, Data Storytelling

Projects

Customer Churn Prediction

github.com

- Developed a customer churn prediction system using Random Forest, and XGBoost on a dataset with 440,834 rows and 11 features, achieving 96 percent accuracy with Ensemble Learning.
- Optimized ML pipeline with EDA, feature engineering, and hyperparameter tuning, reducing training time by 20% while ensuring scalability and robustness.
- **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, XGBoost, FastAPI, Streamlit.

Multi Agentic AI Resume Tailoring

github.com

- Developed a multi-agentic AI system with 4 agents, 4 tasks, async execution, and multi-tool integration using CrewAI to automate resume tailoring and data analysis.
- Implemented structured LLM outputs through a complex agent design, boosting accuracy and cutting manual effort by 70%.

Face Mask Detection

github.com

- Developed a **Face Mask Detection model** using CNN on a dataset of 7500+ images, achieving 92% accuracy through optimized architecture, requirements analysis, and hyperparameter tuning.
- Built an end-to-end deep learning pipeline with image preprocessing, augmentation, batch normalization, and systematic unit testing, **improving model generalization by 15%**.
- **Tech Stack:** Python, TensorFlow, Keras, OpenCV, Matplotlib, NumPy, Pandas.

Experience

Data Science and Analytics Intern, Kerala Ayurveda Ltd. (Hybrid)

Oct 2025 – Present

- Analyzed product market performance across 20+ Ayurvedic SKUs using Python and SQL, uncovering trends that led to a 12% improvement in regional sales strategy.
- Built predictive models (Regression & Time-Series) to forecast product demand with an accuracy rate of 87%, optimizing inventory planning and reducing overstock by 15%.
- Created interactive dashboards in Power BI and R Shiny to visualize customer purchasing behavior, enabling data-driven decisions for marketing and product placement.

Board Member, E-CELL VIT Chennai

Apr 2025 – Present

- Organized 5+ flagship events/workshops, boosting participation by 40%, and managed sponsorships, partnerships, and cross-functional teams to strengthen the startup ecosystem.

Guest Speaker, Machine Learning Workshop, VIT Chennai

Apr 2025

- Delivered a guest lecture to 150 students on core ML concepts and algorithms, earning positive feedback for simplifying complex ideas and inspiring interest in AI applications.

Achievements

- **Certifications:** DeepLearning.AI (*Multi-Agentic AI Systems, Getting Structured Output from LLMs*)
- NDA AIR 4 (2022-1) – Secured All India Rank 4 in NDA entrance exam.
- Branch Rank 3 – Consistently ranked 3rd in department academics for 2 consecutive years.
- 2nd place at V-Medithon Intercollege Hackathon
- 200+ problems solved on LeetCode with 100 days badge.